

Marta Marchiori Manerba

2° Year PhD Student in AI for Society

Born in Mantova, Italy, on 11/3/96 •

Based at Pisa, Italy •

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EDUCATION

University of Pisa, Computer Science Department - **National PhD in AI for Society**

► Supervisors: Dr. Riccardo Guidotti; Prof. Salvatore Ruggieri

Doctoral fellowship on “*Science and technology of interpretable machine learning and explanation of AI-assisted decision making*”

NOVEMBER 2021 - ONGOING

University of Pisa - **Master's Degree** in Digital Humanities: Language Technologies

► Supervisor: Dr. Riccardo Guidotti. Mark: 110/110 with honors

Thesis project on Fairness Auditing in Abusive Language Detection Systems carried out in the field of *Explainability and Fairness in ML*

FEBRUARY 2019 - JULY 2021

University of Pisa - **Bachelor's Degree** in Digital Humanities

► Supervisor: Dr. Maria Claudia Buzzi. Mark: 109/110

Thesis project carried out in the field of *Assistive Technology for Education*: development of a website for the research of tools and ICT (Information and Communications Technology) resources to support cognitive training.

OCTOBER 2015 - FEBRUARY 2019

RESEARCH INTERESTS

NLP; XAI; Fairness in ML;

Algorithmic bias; Algorithmic

Auditing; ML Evaluation; Data

awareness; Digital Discrimination;

Intersectionality; Hate Speech

Detection; Abusive Language

Detection Systems

LANGUAGES

Native speaker: Italian

British School - First Certificate in English (FCE). Passed at Grade A (C1 CEFR Level)

MAY 2019

RESEARCH EXPERIENCE

Supervisor of a Master Thesis in Digital Humanities entitled “Genetic Fairness-Enhancing Data Generation Framework” with Riccardo Guidotti and Martina Cinquini

JANUARY-JULY 2022

Reviewer for LREC Workshop on Perspectivist Approaches to NLP (NLPerspectives)

2022

AIUCD2021 - Assistant moderator during the talk sessions

20-22 JANUARY 2021

Digital Humanities group, Fondazione Bruno Kessler - **Traineeship**

► Fairness analysis of abusive language detection systems detecting unintended models biases with CheckList: creation of synthetic linguistic data to test a range of social prejudices (e.g. sexism, racism and ableism)

OCTOBER - DECEMBER 2020

MEMBERSHIPS

Associazione Italiana di Linguistica Computazionale (Italian Association of Computational Linguistics)

FEBRUARY 2021 - DECEMBER 2021

Associazione per l'Informatica Umanistica e la Cultura Digitale (Association for Digital Humanities and Digital Culture)

JANUARY 2021 - DECEMBER 2021

Supporter of Algorithmic Justice League

2020 - IN PROGRESS

University of Pisa - **Academic Tutor** of the course *Programming and Theoretical Fundamentals*

► Assistance during the laboratory: preparing exercises and assignments, helping students with coding during classes, clarifying basic constructs and concept in Javascript

OCTOBER - DECEMBER 2020

Institute of Computer Science and Telematics, CNR of Pisa - **Traineeship** in *Assistive Technology for Education*

► Mapping and categorization of ICT resources to support cognitive training through the construction of an inventory of tools available on the internet; Implementation of a digital catalogue to make the collection usable

MAY - SEPTEMBER 2018

PUBLICATIONS

Federico Mazzoni, Marta Marchiori Manerba, Martina Cinquini, Riccardo Guidotti, and Salvatore Ruggieri. "FairGen: A Genetic Fairness-Enhancing Data Generation Framework". *Intelligent Data Analysis (IDA)*, 2023

► Currently under review

Marta Marchiori Manerba and Virginia Morini. "Exposing Racial Dialect Bias in Abusive Language Detection: Can Explainability Play a Role?". *ECML PKDD International Workshop on eXplainable Knowledge Discovery in Data Mining (XKDD)*, 2022

► In-person presentation

Marta Marchiori Manerba, Riccardo Guidotti, Lucia Passaro, Salvatore Ruggieri. "Bias Discovery Within Human Raters: A Case Study of the Jigsaw Dataset". *LREC Workshop on Perspectivist Approaches to NLP (NLPerspectives)*, 2022

► In-person presentation

Arianna Muti, Marta Marchiori Manerba, Katerina Korre, and Alberto Barrón-Cedeño. "LeaningTower@LT-EDI-ACL2022: When Hope and Hate Collide". *ACL Workshop on Language Technology for Equality, Diversity and Inclusion (LT-EDI)*, 2022

Marta Marchiori Manerba and Riccardo Guidotti. "Investigating Debiasing Effects on Classification and Explainability". *AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES)*, 2022

► In-person presentation

Marta Marchiori Manerba and Riccardo Guidotti. "FairShades: Fairness Auditing via Explainability in Abusive Language Detection Systems". *IEEE Conference on Cognitive Machine Intelligence (CogMI)*, 2021

► Online presentation, Best Paper Award

Marta Marchiori Manerba and Sara Tonelli. "Fine-Grained Fairness Analysis of Abusive Language Detection Systems with CheckList". *Workshop on Online Abuse and Harms (WOAH!)*, 2021

► Online presentation

DISSEMINATION

“Picture a Scientist”: Organisation of a panel discussion financed by funds for PhD students' scientific initiatives. The event aims to analyse the reasons, factors and motives behind the lower percentage of women choosing to pursue education and careers in STEM fields

2022-2023

Creation of a video for university choice orientation on Digital Humanities and the impacts of AI in the context of the initiative “A Choice for Life”

2022

High school civic education lesson at IIS Cobiauchi in Verbania (online):

“Ethics in AI Systems: Biases in ML Models”

2022

Intervention at a university orientation event for high schools (held at the CS Department in Pisa): “Ethics in AI Systems and the Timnit Gebru case”

2022

DIGITAL ADVOCACY & VOLUNTEERING

Privacy Network - Team member

► Non-profit association that promotes a culture of privacy and responsible use of technology and that advocates for digital rights

NOVEMBER 2022 - IN PROGRESS

KRINO - Vice president and board member (2022)

► Cultural association that organizes panels, workshops and events within the field of Digital Humanities, with the aim of integrating and contaminating sciences and arts

- Active participation in projects of critical divulgation of topics such as: computational thinking, datafication, digital culture, demystification of AI systems, digital art, live coding
- Collaborative and collective writing and editing of posts, articles, newsletters
- Internal training sessions

NOVEMBER 2019 - IN PROGRESS